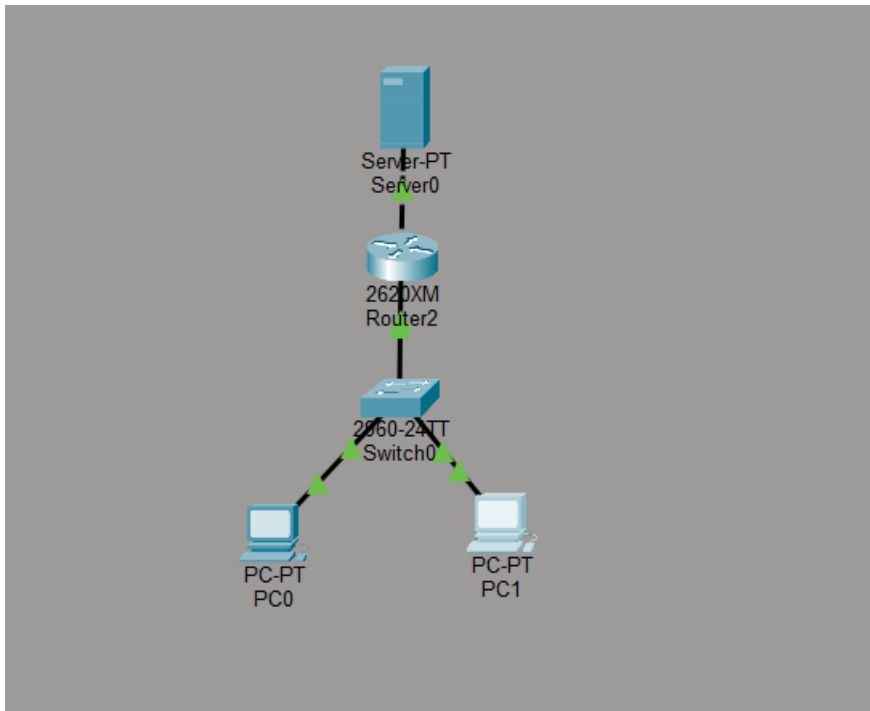


Q-Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

To configure a Cisco router with Network Address Translation (NAT) so that devices on a private Local Area Network (LAN) can communicate with a simulated external server representing the Internet.

Network Topology:



Configure Router Interfaces

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface FastEthernet1/0
Router(config-if)#ip address 203.0.113.1 255.255.255.0
Router(config-if)#ip nat outside
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

Router(config-if)#exit
```

Configure NAT:

```
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat inside source list 1 interface FastEthernet1/0 overload
Router(config)#end
Router#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
```

Configure PCs

PC0

Physical Config Desktop Programming Attributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

☒ DHCP

☒ Static

IPv4 Address192.168.1.2

Subnet Mask255.255.255.0

Default Gateway192.168.1.1

DNS Server0.0.0.0

PC1

Physical Config Desktop Programming Attributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

☒ DHCP

☒ Static

IPv4 Address192.168.1.3

Subnet Mask255.255.255.0

Default Gateway192.168.1.1

DNS Server0.0.0.0

Server0

Physical Config Services Desktop Programming Attributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

☒ DHCP

☒ Static

IPv4 Address203.0.113.2

Subnet Mask255.255.255.0

Default Gateway203.0.113.1

DNS Server0.0.0.0

Enable the **HTTP** service:

1. Open **Server0**.
2. Go to **Services** → **HTTP**.
3. Set **HTTP = ON**.
4. Edit the default web page (index.html) with the following content:

```
<html>
<head><title>Internet Server</title></head>
<body>
<h1>Welcome to the Simulated Internet Server</h1>
<p>NAT Configuration Successful!</p>
</body>
</html>
```

Verification

Test 1: Ping Router

From **PC0**:

ping 192.168.1.1

Result: Successful

Test 2: Ping External Server

From **PC0**:

ping 203.0.113.2

Result: Successful

Test 3: Access External Server

1. Open **PC0** → **Desktop** → **Web Browser**.
2. Enter:

<http://203.0.113.2>



Test 4: Verify NAT Translation

On the router, execute:

show ip nat translations

Output:

```
Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
tcp  203.0.113.1:1025    192.168.1.2:1025  203.0.113.2:80    203.0.113.2:80
Router#
```

Network Address Translation (NAT) was successfully configured on the Cisco 2620XM router. The router translated private IP addresses from the LAN (192.168.1.0/24) to its public interface (203.0.113.1), allowing LAN devices to access the simulated external server (203.0.113.2). Successful ping responses, HTTP access, and NAT translation entries verified the correct operation of NAT.